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**TASK
3**

**CEF 440: INTERNET PROGRAMMING AND MOBILE
PRORAMMING**

TASK1: REQUIREMENTS ANALYSIS

GROUP 19

**Design and Implementation of a Mobile App for Collection of User Experience Data
from Mobile Network Subscribers**

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TABLE OF CONTENT

1. Introduction	3
2. Requirement Analysis	4
2.1 Review and Analysis of Gathered Requirements	4
2.2 Identification of Inconsistencies and Missing Information	4
2.3 Requirement Prioritization	5
2.4 Requirement Classification.....	5
2.5 Validation of Requirements with Stakeholders	6
3. Software Requirements Specification (SRS)	6
3.1 Purpose of the System	6
3.2 Scope of the System	7
3.3 Stakeholders and Users.....	7
3.4 Functional Requirements	7
3.5 Non-Functional Requirements.....	8
3.6 System Features	9
3.7 Hardware and Software Requirements	9
3.8 Constraints and Assumptions	9
3.9 Security Requirements.....	9
3.10 Future Improvements.....	10
4. Conclusion.....	11

1. Introduction

The increasing dependence on mobile internet services has significantly raised user expectations regarding network quality, reliability, and overall service performance. In Cameroon, many mobile subscribers experience poor connectivity, unstable bandwidth, high latency, call interruptions, and inconsistent browsing experiences. Despite the existence of technical monitoring systems used by network operators, these systems often fail to fully capture the real Quality of Experience (QoE) perceived by users.

This project focuses on the design and implementation of a mobile application capable of collecting both subjective and objective QoE data from mobile network subscribers in real time. The application continuously monitors network and device parameters while also allowing users to provide satisfaction ratings and feedback regarding their experiences.

The requirement analysis phase was conducted to identify, analyze, validate, and organize all system requirements necessary for successful development. The analysis was based on the requirement gathering activities previously conducted using surveys, interviews, brainstorming sessions, and Speedtest application results.

2. Requirement Analysis

2.1 Review and Analysis of Gathered Requirements

During the requirement gathering phase, we collected information from mobile subscribers through surveys and observations. We also analyzed objective network metrics obtained using the Speedtest application. Existing platforms such as OpenSignal and NetMonster were reviewed to understand how similar systems operate.

The gathered requirements were analyzed based on the following factors:

- **Completeness**

We verified whether the collected requirements fully addressed user needs and project objectives. Users consistently requested features such as automatic network monitoring, background execution, battery efficiency, and real-time feedback collection.

- **Clarity**

Requirements were rewritten in simple and understandable language to ensure that both technical and non-technical stakeholders could understand them easily.

- **Technical Feasibility**

We evaluated whether the requested functionalities could be implemented using Android APIs and available technologies. Features such as signal strength monitoring, latency measurement, location tracking, and background execution were confirmed to be technically feasible.

- **Dependency Relationships**

Some requirements depend on others. For example:

- Real-time analytics depend on continuous data collection.
- QoE reporting depends on successful storage of network metrics and user feedback.
- User notifications depend on background services running properly.

2.2 Identification of Inconsistencies and Missing Information

After reviewing the gathered data, some inconsistencies and missing information were identified.

- **Inconsistencies**

Some users reported acceptable browsing experiences even when Speedtest results indicated low bandwidth or high latency. This showed that user perception may differ from actual network measurements.

• **Missing Information**

Certain requirements were not initially clear, including:

- Preferred frequency for feedback notifications
- Acceptable battery consumption levels
- Offline data synchronization requirements
- Device compatibility requirements

These issues were clarified through additional analysis and follow-up discussions.

2.3 Requirement Prioritization

Requirements were prioritized according to importance, feasibility, and impact on system objectives.

High Priority Requirements:

- Continuous background monitoring
- Automatic collection of network metrics
- User satisfaction feedback collection
- Secure storage of collected data
- Timestamp and GPS location recording

Medium Priority Requirements:

- Data visualization dashboard
- Offline data synchronization
- Notification and alert system

Low Priority Requirements:

- Social sharing functionality
- Multi-language support
- Personalized themes

Prioritization helps developers focus on the most critical functionalities first.

2.4 Requirement Classification

The system requirements were classified into functional and non-functional requirements.

Functional Requirements:

- The application must collect network metrics automatically.
- The application must allow users to submit feedback and satisfaction ratings.
- The application must store timestamps and location data.
- The application must synchronize collected data with the server.
- The application must generate QoE reports for analysis.

Non-Functional Requirements:

- The system must be secure and protect user privacy.
- The application must consume minimal battery power.
- The system must provide high reliability and availability.
- The application must be user-friendly and easy to use.
- The system must support scalability for large numbers of users.

2.5 Validation of Requirements with Stakeholders

The identified requirements were validated with stakeholders to ensure that they align with actual user needs and business objectives.

Validation involved:

- Reviewing survey results with team members
- Comparing findings with existing platforms
- Evaluating requirements against project objectives
- Considering the expectations of mobile network operators such as MTN Cameroon, Orange Cameroon, and Camtel

Stakeholders confirmed that:

- Real-time QoE monitoring is necessary
- User privacy must be prioritized
- Automated data collection is more reliable than manual reporting
- Lightweight applications are preferred by users

3. Software Requirements Specification (SRS)

The Software Requirements Specification (SRS) document defines the complete behavior and functionality of the proposed QoE mobile application. It serves as a communication tool between developers, stakeholders, testers, and project managers. The SRS ensures that all parties clearly understand the system objectives, requirements, constraints, and expected outcomes.

3.1 Purpose of the System

The purpose of the system is to develop a mobile application capable of collecting, monitoring, storing, and analyzing Quality of Experience (QoE) data from mobile network subscribers in real time.

The application combines:

- Subjective feedback from users

- Objective network performance metrics

The collected data will support:

- Network optimization
- Service quality evaluation
- Customer satisfaction improvement
- Decision-making by network operators

3.2 Scope of the System

The system will operate on Android mobile devices and perform the following functions:

- Run continuously in the background
- Collect network metrics automatically
- Prompt users periodically for feedback
- Store and transmit data securely
- Generate QoE analytics and reports

The application focuses on mobile network performance monitoring within Cameroon.

3.3 Stakeholders and Users

Primary Users:

- Mobile subscribers
- Network analysts
- System administrators

Secondary Stakeholders:

- Mobile network operators
- Researchers
- Regulatory authorities

Each stakeholder interacts differently with the system:

- Subscribers provide feedback
- Analysts review reports
- Operators use data for optimization

3.4 Functional Requirements

1. User Authentication

- The system must allow secure user login and registration.

2. Background Monitoring

- The application must continuously collect network metrics while running in the background.

3. Feedback Collection

- Users must be able to rate their network experience.

4. Network Metrics Collection

The application must collect:

- Signal strength
- Network type
- Download speed
- Upload speed
- Latency
- Packet loss
- Jitter
- Timestamp
- GPS location

5. Data Synchronization

- The application must send collected data to a backend server.

6. Report Generation

- The system must generate reports and analytics dashboards.

3.5 Non-Functional Requirements

1. Performance

- The application must respond quickly and operate efficiently.

2. Security

- User data must be encrypted and protected.

3. Reliability

- The system must operate continuously without crashes.

4. Usability

- The interface must be simple and intuitive.

5. Scalability

- The system must support large numbers of users.

6. Maintainability

- The system should be easy to update and maintain.

3.6 System Features

Main features of the application include:

- Real-time QoE monitoring
- Background execution
- User feedback forms
- Speedtest integration
- Interactive dashboard
- Notification system
- Analytics and visualization tools

3.7 Hardware and Software Requirements

Hardware Requirements:

- Android smartphone
- GPS-enabled device
- Internet connectivity

Software Requirements:

- Android Studio
- Java/Kotlin
- Firebase/MySQL
- REST APIs
- Node.js backend

3.8 Constraints and Assumptions

Constraints:

- Internet access is required for synchronization.
- Android permissions are necessary for location and network monitoring.

Assumptions:

- Users will allow background execution.
- Mobile operators will benefit from collected QoE data.
- Users will provide honest feedback.

3.9 Security Requirements

To protect user information:

- Data transmission must use encryption.
- Authentication mechanisms must be implemented.

- Personal information must remain confidential.
- Access to reports must be restricted to authorized users only.

3.10 Future Improvements

Possible future improvements include:

- AI-based QoE prediction
- Cross-platform support (iOS)
- Integration with telecom operator systems
- Advanced machine learning analytics
- Real-time network issue alerts

4. Conclusion

The requirement analysis phase provided a detailed understanding of user expectations, system objectives, and technical feasibility for the QoE mobile application project. Through surveys, Speedtest results, and analysis of existing platforms, we identified the major challenges faced by mobile network subscribers in Cameroon. The Software Requirements Specification document clearly defines the system functionalities, constraints, and performance expectations in a manner understandable to both developers and stakeholders. This phase establishes a strong foundation for the next stages of system modelling, design, and implementation.